

Muhammad Khan

ML Engineer

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[Github](#)

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PROJECTS

Movie Recommendation System [Live](#)

- ▶ Built a content-based recommendation engine using TF-IDF vectorization on movie metadata (genres, cast, keywords, overview).
- ▶ Computed pairwise cosine similarity matrix to rank and retrieve top-N most similar movies for any given title.
- ▶ Deployed an interactive Streamlit web app allowing users to search for any movie and receive instant recommendations.
- ▶ Preprocessed and cleaned a dataset of 5,000+ movies, handling missing values and normalizing text features.

Gold Price Predictor [Live](#)

- ▶ Developed a regression model to forecast gold prices using macroeconomic indicators (USD index, S&P 500, oil prices, silver).
- ▶ Performed extensive EDA with Matplotlib and Seaborn, identifying key correlations and feature importance.
- ▶ Trained and tuned a Random Forest Regressor achieving high predictive accuracy with cross-validation.

Titanic Survival Prediction [Live](#)

- ▶ Conducted thorough feature engineering: imputed missing ages, encoded categorical variables (sex, embarked), and engineered family-size features.
- ▶ Compared multiple classifiers (Logistic Regression, Decision Tree, Random Forest) using accuracy, precision, recall, and F1-score.
- ▶ Achieved Highest accuracy on the test set using an optimized Logistic Regression model.

Built an interactive Streamlit dashboard to visualize survival distributions and make predictions for custom passenger profiles

EDUCATION

University of Punjab, Lahore— BSCS

2024-2028 (currently 4th semester)

Reader College, Sargodha— ICS

2022-2024

SUMMARY

I work as a Data Scientist and Machine Learning Engineer. My specialty is creating machine learning systems. I do this by building pipelines that get the data ready for use. I am skilled at using tools like FastAPI and Streamlit. These tools help me take data and turn it into insights that businesses.

SKILLS

Language: Python

Visualization: NumPy, Pandas, Matplotlib, Seaborn

Machine Learning: Scikit-Learn

• ML Algorithms • Feature Engineering • Model Evaluation

NLP: NLTK • TF-IDF Vectorization

• Bag of Words • Text Preprocessing
• Cosine Similarity

Web Frameworks: FastAPI • Streamlit